

Collections

1) Create an Employee class with following properties

EmployeeName as string

EmployeeID as int

Salary as double

Create another class called EmployeeDAL which has the following methods.

1. Create an ArrayList within this class

2. Define the below methods

a. public bool AddEmployee (Employee e)

b. public bool DeleteEmployee (int id)

c. public string Search Employee (int id): Returns the employee name if found

else return null

d. public Employee [] GetAllEmployeesistAll()

Use an ArrayList to store the employees

Write a test program to test all the functionalities.

```
using System;
```

```
using System.Collections;
```

```
class Employee
```

```
{
```

```
    public string EmployeeName { get; set; }
```

```
    public int EmployeeID { get; set; }
```

```
    public double Salary { get; set; }
```

```
    public Employee(string name, int id, double salary)
```

```
    {
```

```
        EmployeeName = name;
```

```

        EmployeeID = id;
        Salary = salary;
    }
}
class EmployeeDAL
{
    ArrayList employees = new ArrayList();
    public bool AddEmployee(Employee e)
    {
        employees.Add(e);
        return true;
    }
    public bool DeleteEmployee(int id)
    {
        foreach (Employee emp in employees)
        {
            if (emp.EmployeeID == id)
            {
                employees.Remove(emp);
                return true;
            }
        }
        return false;
    }
    public string SearchEmployee(int id)
    {
        foreach (Employee emp in employees)
        {
            if (emp.EmployeeID == id)
                return emp.EmployeeName;
        }
    }
}

```

```

    }
    return null;
}
public Employee[] GetAllEmployees()
{
    Employee[] arr = new Employee[employees.Count];
    for (int i = 0; i < employees.Count; i++)
    {
        arr[i] = (Employee)employees[i];
    }
    return arr;
}
}
class Program
{
    static void Main()
    {
        EmployeeDAL dal = new EmployeeDAL();
        Console.Write("How many employees to add? ");
        int n = Convert.ToInt32(Console.ReadLine());
        for (int i = 0; i < n; i++)
        {
            Console.WriteLine($"Enter details for Employee {i + 1}");
            Console.Write("Employee ID: ");
            int id = Convert.ToInt32(Console.ReadLine());
            Console.Write("Employee Name: ");
            string name = Console.ReadLine();
            Console.Write("Salary: ");
            double salary = Convert.ToDouble(Console.ReadLine());
            dal.AddEmployee(new Employee(name, id, salary));
        }
    }
}

```

```

    }
    Console.WriteLine("\nEnter Employee ID to Search: ");
    int searchId = Convert.ToInt32(Console.ReadLine());
    string result = dal.SearchEmployee(searchId);
    if (result != null)
        Console.WriteLine("Employee Found: " + result);
    else
        Console.WriteLine("Employee Not Found");
    Console.WriteLine("\nEnter Employee ID to Delete: ");
    int deleteId = Convert.ToInt32(Console.ReadLine());
    if (dal.DeleteEmployee(deleteId))
        Console.WriteLine("Employee Deleted Successfully");
    else
        Console.WriteLine("Employee Not Found");
    Console.WriteLine("\nAll Employees:");
    Employee[] list = dal.GetAllEmployees();
    foreach (Employee emp in list)
    {
        Console.WriteLine(emp.EmployeeID + " - " + emp.EmployeeName + " - " +
emp.Salary);
    }
    Console.ReadLine();
}
}

```

OUTPUT

```
PS C:\Users\Nandu\OneDrive\Desktop\trining\wipro\Datatypes> dotnet run
How many employees to add? 2
```

```
Enter details for Employee 1
```

```
Employee ID: 101
```

```
Employee Name: ram
```

```
Salary: 50000
```

```
Enter details for Employee 2
```

```
Employee ID: 102
```

```
Employee Name: krishna
```

```
Salary: 55000
```

```
Enter Employee ID to Search: 102
```

```
Employee Found: krishna
```

```
Enter Employee ID to Delete: 103
```

```
Employee Not Found
```

```
All Employees:
```

```
101 - ram - 50000
```

```
102 - krishna - 55000
```

```
█
```